



Zhengyang Wang (Alvin)

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EDUCATION

- **Beijing Normal-Hong Kong Baptist University (BNBU)** Zhuhai, China
B.Sc. (Honours) in Computer Science and Technology Sep 2023 – Jun 2027 (expected)
 - **CGPA:** 3.24/4.00 | **Medium of instruction:** English
 - **Selected Coursework:** Probability and Statistics (A); 3D Computer Vision (A-); Data Analytics for Business (A-); Machine Learning (B+); Neural Networks and Deep Learning (B+); Data Structures and Algorithms (B+)

RESEARCH EXPERIENCE

- **Research and Development of Third-Generation Intelligent Arrow Selection Equipment and Systems** Zhuhai, China
Research Assistant / Digital Sports Research Centre, BNBU May 2025 – May 2026
(Supervisors: Huaxiong Huang, DKU; Lidong Fang, SUFE)
 - Curated and manually validated a multimodal dataset linking arrow measurements, shooting trials, target images, impact coordinates and ring scores to assess shot dispersion and stability across rounds for data-driven arrow selection
 - Extracted launch velocity and angle from high-speed video to analyse how shot kinematics relate to impact patterns
 - Led the fine-tuning and integration of SAM 3.1 and YOLO11s for archery target segmentation and arrow impact localisation, achieving 92.58% precision, 88.51% recall, and 90.50% F1 via offline temporal fusion
 - **Manuscript:** Co-author of *Arrow Selection for Elite Series Arrows: Data Driven Archery Arrow Selection*, under peer review at *Sports Engineering*
- **MetroHeight: Mobile RGB-D Cargo Height Estimation** Apr 2026 – Present
(First author / Supervisor: Xiubin Fan, BNBU)
 - Built an iOS capture app and FastAPI backend for cargo-height inspection of delivery trolleys in metro stations
 - Aligned camera intrinsics to depth-map resolution, filtered invalid and low-confidence depth samples, and used DBSCAN clustering to isolate cargo geometry before height estimation
 - Implemented SAM 3 segmentation and gravity-aligned 3D reconstruction, combining local RANSAC floor fitting with robust cargo-top estimation to measure floor-to-cargo height while excluding trolley structures
 - Evaluated 194 capture records across four trolley categories at Hong Kong MTR's Wong Tai Sin Station against laser reference measurements, achieving 3.13 cm MAE and 4.81 cm RMSE
 - **Manuscript:** First author of *MetroHeight: Smartphone RGB-D Measurement of Cargo Height in Metro Stations*; manuscript in preparation for submission to *ACM SenSys 2027*
- **Spatiotemporal Flood Forecasting and Hydrological Modelling** Sep 2025 – Sep 2026
Research under the Guangdong Major Project of Basic and Applied Basic Research:
High-Resolution Simulation and Prediction of the Land Surface Environment (Project No. 2021B0301030007)
(Supervisors: Huaxiong Huang, DKU; Zijian Wang, BNBU)
 - Developed a flow-direction-aware river-discharge forecasting method combining hydrological time series with spatial river-network graphs to represent downstream hydrological propagation
 - Integrated ERA5-Land reanalysis, hydrometric observations from 75 gauges across 15 rivers, and static geospatial rasters into river-specific 29-channel grids, with water levels forward-filled for up to seven days
 - Implemented event-aware raster-Transformer training with river-balanced regression and high-flow underprediction penalties, and assessed three-seed ensembles using chronological splits and validation-based checkpoint selection
 - **Conference abstract:** Second author of *A Spatiotemporal Hybrid Framework for Regional Flood Forecasting*; accepted for poster presentation at WRE 2026, Seoul, Dec 2026

- **Visual Metaphor and Multimodal Reasoning** Apr 2026 – May 2026
 - Led MetaVision-DB data construction, covering 800 images, four visual–semantic relations and three communication games, with concept–image alignment, schema validation and manual quality review
 - Co-developed MACH, a dual-block VLM agent framework combining contrastive visual-semantic reasoning and strategy-conditioned metaphor generation with structured memory, theory-derived knowledge and simulated teammate and opponent responses
 - Evaluated MACH in Visual Undercover, where three MACH civilian agents achieved a 97.2% win rate against two Raw-VLM undercover agents using Gemini 3.1 Pro
 - **Manuscript:** Third author of *Orchestrating Secrets: Visual Metaphor Generation and Reasoning in Multi-Agent Communication Games*; in revision for ARR resubmission
- **MapAgent: Indoor Localisation and Spatial Reasoning** Jul 2026 – Present
(Final Year Project (FYP) / Supervisor: Xiubin Fan, BNBU)
 - Developed a Qwen3-based localisation agent that reasons over BLE, motion and map evidence to select candidate locations, with independent verification and deterministic fallback
 - Evaluated the system across six sites and 41 trajectories, obtaining a trajectory-averaged localisation error of 3.317 m
- **Cardiac Segmentation and Spatiotemporal Modelling** Apr 2026 – Aug 2026
 - Developed ST-Net for $2D + t$ cine MRI segmentation with a bottleneck Temporal Attention Gate and second-order cyclic regularisation of area and centroid trajectories under ED/ES-only supervision
 - Benchmarked seven model and loss configurations on 50 ACDC cine MRI cases using five-fold patient-level cross-validation
 - **Manuscript:** Third author of *ST-Net: Temporal Recalibration for Cardiac Cine Segmentation under ED/ES-only Labels*; submitted to AISTATS 2027
- **Hydrological Constraints on Post-Fire Vegetation Recovery** Jun 2026 – Sep 2026
 - Analysed modelled soil-water saturation and MODIS NDVI across equal six-year pre- and post-fire periods, using the 2019–2020 Gaspers Mountain megafire as a natural experiment to assess hydrological constraints on vegetation recovery
 - Quantified an increase in positive soil-water–NDVI trend concordance from 37.8% before fire to 95.3% during 2020–2025 recovery, while finding no positive coupling in year-to-year changes
 - **Conference abstract:** Second author of *Long-term hydrological support constrains vegetation recovery after major disturbance*; accepted for in-person poster presentation at BES Annual Meeting 2026

INTERNSHIPS & OPEN SOURCE

- **Tencent Rhino-Bird Open Source Talent Program** Aug 2026
 - Delivered KuiklyLoadingKit, a Kotlin Multiplatform loading component for Tencent KuiklyUI with a declarative DSL and Android/iOS demos; completion confirmed by a project collaborator
- **Bank of Guangzhou, Hengqin Branch** Zhuhai, China
Corporate Finance Department Intern Jan 2025 – Feb 2025
 - Organised corporate lending and credit records in Excel, supporting document review and information management
- **Beijing Normal-Hong Kong Baptist University (BNBU)** Zhuhai, China
Finance Office Intern — Work-study Programme Sep 2024 – Dec 2024
 - Supported expense reimbursement, voucher organisation and document filing

HONOURS & AWARDS

- **Dean's List** (five semesters), BNBU – 2023–2026
- **Excellence Award**, ICBC Cup University FinTech Innovation Competition – 2023
- **Guangdong Provincial Excellence Award**, China Undergraduate Mathematical Contest in Modeling (CUMCM) – Oct 2025
- **Bronze Award of Servant Leadership**, Student Leadership Path Programme, BNBU – May 2026

LEADERSHIP & ACTIVITIES

- **Student Union Staff**, Student Rights & Welfare Department, BNBU – 2023–2024
- **Organisation Officer**, Communist Youth League Class Branch, BNBU – 2023–2024
- **Co-founder**, BNBU Volleyball Club

TECHNICAL SKILLS & LANGUAGES

Technical Skills: Python, SQL, Swift, Kotlin, FastAPI, PyTorch, OpenCV, Git, LaTeX

Languages: Mandarin (Native); English (IELTS Academic Overall 6.5; CET-4 and CET-6)